

Cabin Crew Scheduling

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The Problem

- 21% of operating expenses (second only to fuel)
- 45.8% of airline staff
- Bad schedules have consequences – missed duty limits stranded passengers for hours

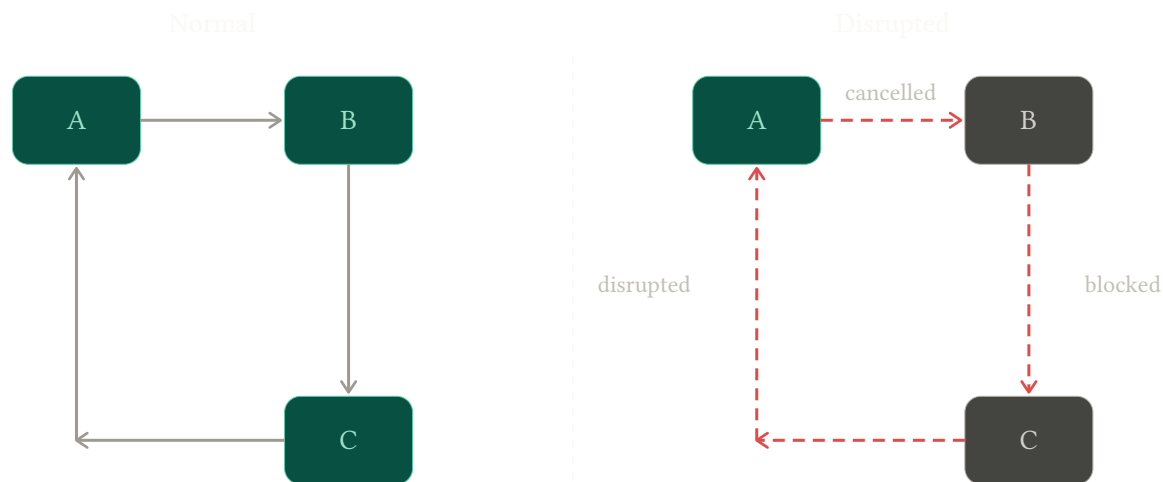
- 21% of operating expenses (second only to fuel)
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Cut costs while keeping every flight covered

- Cross-qualified across multiple aircraft types

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- Categorised into multiple classes according to skills and experiences
- One flight delay/ cancellation may disturb future scheduling



Assumptions

- The flight schedule, fleet assignment and aircraft routing have already been solved upstream
- The planning horizon is one week

Original Paper Formulations

R Set of cabin crew classes

F Set of scheduled flights

B Set of home bases

T Set of feasible **team** pairings for cabin crews

P_r Set of feasible **individual** pairings for Class r **available** crew.

P_r^e Set of feasible **individual** pairings for Class r **extra** cabin crews

Each pairing must start and end at the crew's home base

cost_t TAFB (Time Away From Base) cost of team pairing $t \in T$

$\text{cost}_{rp}^{\text{avail}}$ TAFB cost of individual pairing $p \in P_r$ for Class r
available cabin crew

$\text{cost}_{rp}^{\text{extra}}$ TAFB cost of individual pairing $p \in P_r^e$ for Class r extra
cabin crew

covers_{ft} $= \begin{cases} 1 & \text{if team pairing } t \text{ covers flight } f \in F \\ 0 & \text{otherwise} \end{cases}$

$\text{covers}_{frp}^{\text{avail}}$ $= \begin{cases} 1 & \text{if individual pairing } p \in P_r \text{ covers flight } f \in F \\ 0 & \text{otherwise} \end{cases}$

$\text{covers}_{frp}^{\text{extra}}$ $= \begin{cases} 1 & \text{if extra individual pairing } p \in P_r^e \text{ covers flight } f \in F \\ 0 & \text{otherwise} \end{cases}$

req_{rf}	Number of Class r cabin crews required by flight f
avail_r	Total number of available Class r cabin crews
μ	Unit substitution penalty cost; satisfies $\text{cost}_{rp}^{\text{avail}}, \text{cost}_{rp}^{\text{extra}} \ll \mu \ll M$
M	penalty cost for employing extra cabin crew

$$\text{select}_t = \begin{cases} 1 & \text{if team pairing } t \text{ is selected} \\ 0 & \text{otherwise} \end{cases}$$

$\text{assign}_{rp} \in \mathbb{Z}_{\geq 0}$ Number of times individual pairing $p \in P_r$ is used for Class r available cabin crew

$\text{extra}_{rp} \in \mathbb{Z}_{\geq 0}$ Number of times individual pairing $p \in P_r^e$ is used for Class r extra cabin crew

$\text{sub}_{rf} \in \mathbb{Z}_{\geq 0}$ Number of times Class r cabin crew is substituted by another class on flight f

Traditional Cabin Crew Pairing Problem (TCCPP)

- Minimise the total TAFB pairing cost

$$\min \sum_{t \in T} \text{cost}_t \cdot \text{select}_t$$

Traditional Cabin Crew Pairing Problem (TCCPP)

Original Paper Formulations

- Each flight f must be covered by at least one selected team pairing

$$\sum_{t \in T} \text{covers}_{ft} \cdot \text{select}_t \geq 1, \quad \forall f \in F$$

Pairing Problem with Availability and Controlled Crew Substitution

$$\min \underbrace{\sum_{r \in R} \sum_{p \in P_r} \text{cost}_{rp}^{\text{avail}} \cdot \text{assign}_{rp}}_{\text{available crew cost}}$$

Multi-class Individual Cabin Crew Pairing Problem with Availability and Controlled Crew Substitution

Original Paper Formulations

$$\begin{aligned} \min \quad & \underbrace{\sum_{r \in R} \sum_{p \in P_r} \text{cost}_{rp}^{\text{avail}} \cdot \text{assign}_{rp}}_{\text{available crew cost}} \\ & + \underbrace{\sum_{r \in R} \sum_{f \in F} \mu \cdot \text{sub}_{rf}}_{\text{substitution penalty}} \end{aligned}$$

Pairing Problem with Availability and Controlled Crew Substitution

$$\begin{aligned}
 \min \quad & \underbrace{\sum_{r \in R} \sum_{p \in P_r} \text{cost}_{rp}^{\text{avail}} \cdot \text{assign}_{rp}}_{\text{available crew cost}} \\
 & + \underbrace{\sum_{r \in R} \sum_{f \in F} \mu \cdot \text{sub}_{rf}}_{\text{substitution penalty}} \\
 & + \underbrace{\sum_{r \in R} \sum_{p \in P_r^e} (\text{cost}_{rp}^{\text{extra}} + M) \cdot \text{extra}_{rp}}_{\text{extra crew cost}}
 \end{aligned}$$

Total satisfaction constraint

the total number of cabin crews of all classes assigned to flight f must meet the total demand across all classes.

$$\sum_{r \in R} \sum_{p \in P_r} \text{covers}_{f r p}^{\text{avail}} \cdot \text{assign}_{r p} + \sum_{r \in R} \sum_{p \in P_r^e} \text{covers}_{f r p}^{\text{extra}} \cdot \text{extra}_{r p} \geq \sum_{r \in R} \text{req}_{r f}, \quad \forall f \in F$$

Minimum satisfaction constraint

At least one qualified crew member from each class must be assigned to every flight f .

$$\sum_{p \in P_r} \text{covers}_{f r p}^{\text{avail}} \cdot \text{assign}_{r p} + \sum_{p \in P_r^e} \text{covers}_{f r p}^{\text{extra}} \cdot \text{extra}_{r p} \geq 1$$
$$\forall f \in F, \forall r \in R$$

Substitution recording constraint

Tracks the number of times Class r is substituted on flight f

$$\sum_{p \in P_r} \text{covers}_{f r p}^{\text{avail}} \cdot \text{assign}_{r p} + \sum_{p \in P_r^e} \text{covers}_{f r p}^{\text{extra}} \cdot \text{extra}_{r p} + \text{sub}_{r f} \geq \text{req}_{r f}$$
$$\forall f \in F, \forall r \in R$$

Crew availability constraint

$$\sum_{p \in P_r} \text{assign}_{rp} \leq \text{avail}_r, \quad \forall r \in R$$

Non-negativity

$$\text{assign}_{rp} \in \mathbb{Z}_{\geq 0}, \quad \forall r \in R, \forall p \in P_r$$

$$\text{extra}_{rp} \in \mathbb{Z}_{\geq 0}, \quad \forall r \in R, \forall p \in P_r^e$$

$$\text{sub}_{rf} \in \mathbb{Z}_{\geq 0}, \quad \forall f \in F, \forall r \in R$$

$$\min \sum_{p \in P_r} \text{cost}_{rp}^{\text{avail}} \cdot \text{assign}_{rp} + \sum_{p \in P_r^e} (\text{cost}_{rp}^{\text{extra}} + M) \cdot \text{extra}_{rp}$$

For each $r \in R$, subject to:

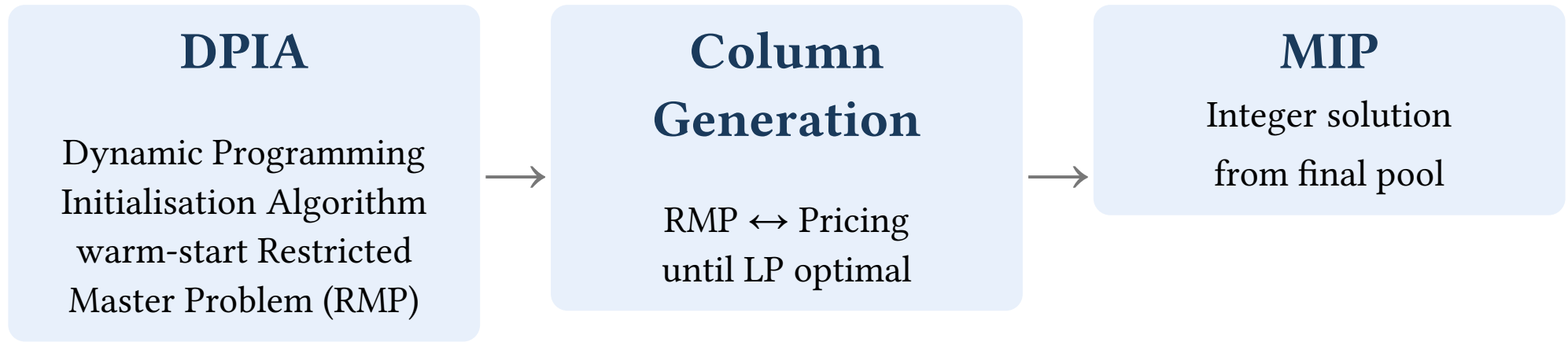
$$\sum_{p \in P_r} \text{covers}_{frp}^{\text{avail}} \cdot \text{assign}_{rp} + \sum_{p \in P_r^e} \text{covers}_{frp}^{\text{extra}} \cdot \text{extra}_{rp} \geq \text{req}_{rf}, \quad \forall f \in F$$

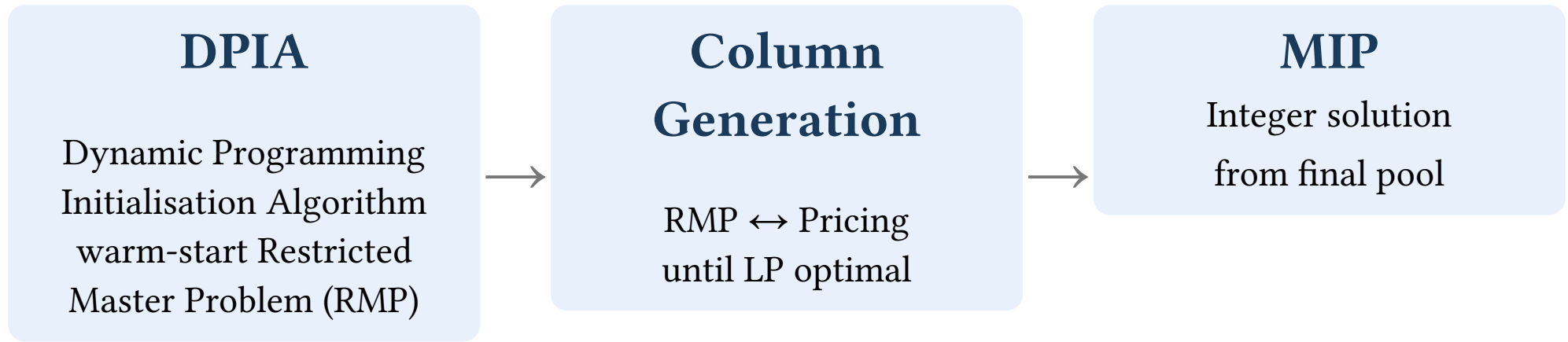
$$\sum_{p \in P_r} \text{assign}_{rp} \leq \text{avail}_r$$

$$\text{assign}_{rp} \in \mathbb{Z}_{\geq 0}, \quad \forall p \in P_r$$

$$\text{extra}_{rp} \in \mathbb{Z}_{\geq 0}, \quad \forall p \in P_r^e$$

Original Paper Approach





For large instances: replace CG + MIP with Genetic Algorithm.

Provides the warm-start that makes CG tractable from the first iteration.

- Quickly generates a small set of legal pairings covering every flight
- Uses extra crew variables — feasibility always guaranteed
- No crew availability constraints at this stage

RMP and PP alternate till no improving column exists $\rightarrow$ LP optimality

RMP

- LP relaxation of the full model
- Solved by **Simplex**
- Outputs **dual prices** per constraint

PP

- Finds pairings with **negative reduced cost**
- Resource Constrained Shortest Path (RCSPP)
- Solved via **DP labelling** on duty network

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- MIP is solved over only those columns
- Recovers integer (assignable) pairings from the LP solution
- global optimality is not guaranteed

Original Paper Limitations

Method

Restricting MIP to the last RMP pool is fast, but may miss integer-optimal solutions that were never generated.

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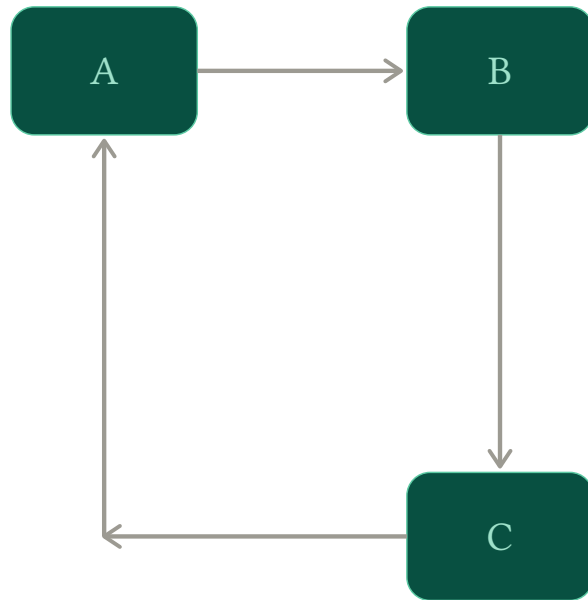
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Substitution Order

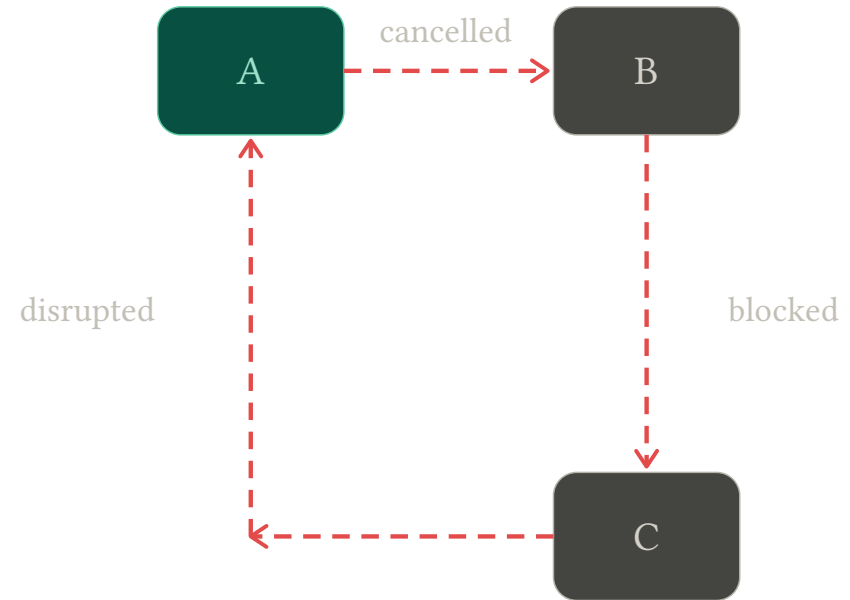
- “All classes can substitute any other class in either direction”

Flight Delay and Cancellations

Normal



Disrupted



Extensions & Plan of Attack

- Compute costs for
 - Flight hours
 - Layovers
 - Deadhead

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- Restricting only senior crew can substitute general cabin mate but not the other way around

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 - Flight hours
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 - Deadhead
- Restricting only senior crew can substitute general cabin mate but not the other way around
- Update schedules upon flight delay and cancellations

- Replicate MICCPP-ACCS

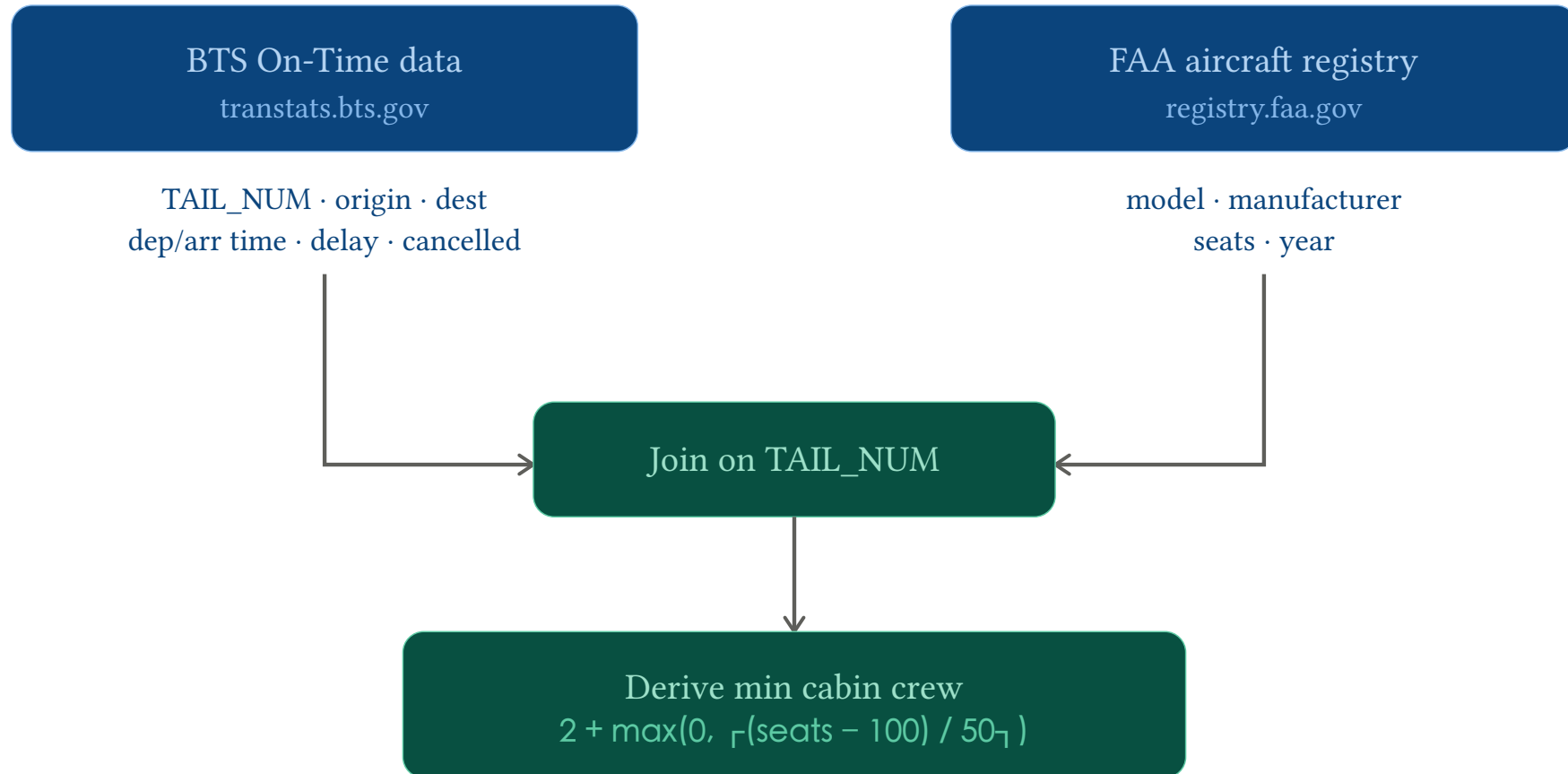
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- Domain independent dynamic programming
- D-star algo?

Data availability

The data that has been used is confidential.



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So is the model.

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So is the model.

So is this slide.

Reference

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- <https://registry.faa.gov/database/ReleasableAircraft.zip>
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